THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** autotrophic	Plants make their own food.
2	**(d)** neuron	Neuron.
3	**(c)** produce new individuals like themselves	Basic definition.
4	**(c)** heredity and variation	Genetics deals with inheritance.
5	**(c)** bile	Bile juice.
6	**(a)** insulin	Insulin.
7	**(c)** pollen grains	Anther is the pollen-making part.
8	**(a)** 46	23 pairs.
9	**(c)** four	Two atria and two ventricles.
10	**(c)** phototropism	Phototropism.
11	**(a)** sperm	Sperm cells are male gametes.
12	**(d)** Colour of the eyes	Controlled by genes.
13	**(d)** CO_2 and water	CO_2 + H_2O.
14	**(c)** axon	Axon.
15	**(b)** binary fission	Unicellular organisms divide.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Process of obtaining and using food (1); Autotrophic and heterotrophic (1).	1+1
17	Nerve cell (1); Dendrite, cell body, axon (1).	1+1
18	Production of new individuals of the same kind (2).	2
19	Gregor Mendel (2).	2
20	Rhythmic contraction of muscles of the alimentary canal (1); Pushes food forward (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Carbon dioxide from air (1); Water from soil (1); Light from Sun, chlorophyll in chloroplast (1).	1+1+1
22	Voluntary: brain-controlled (1); Reflex: spinal cord (1); Reflex is faster (1).	1+1+1
23	Binary fission: Amoeba (1); Budding: Hydra (1); Spore formation: Rhizopus (1).	1+1+1
24	Seed shape (round or wrinkled) (1); Seed colour (yellow or green) (1); Plant height (tall or dwarf) (1).	1+1+1
	OR: Absorbs water (1); Stores undigested food (1); Egestion (1).	1+1+1
25	Insulin: pancreas (1); Thyroxine: thyroid (1); Adrenaline: adrenal (1).	1+1+1
	OR: Ovary (1); Oviduct (1); Uterus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Takes place in chloroplast (1); Chlorophyll absorbs light (1); Water splits (1); CO_2 is reduced to glucose (1); $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (1).	1+1+1+1+1
	OR: Dendrite receives information (1); Cell body has nucleus (1); Axon carries impulse (1); Nerve endings pass impulse (1); Synapse between neurons (1).	1+1+1+1+1
27	Amoeba is a single cell with a nucleus (1); The nucleus divides into two (1); Cytoplasm constricts in the middle (1); Two daughter cells separate (1); Each grows into a new Amoeba (1).	1+1+1+1+1

	OR: Parents TT x tt (1); F1 is Tt, all tall (1); Gametes of F1 are T and t (1); F2 has TT, Tt, Tt, tt (1); Ratio is 3 tall : 1 dwarf (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** garden pea	Pisum sativum.
2	**(c)** liver	Liver.
3	**(d)** thyroid	Thyroid.
4	**(a)** stigma	It receives pollen.
5	**(b)** XY	One X and one Y.
6	**(c)** arteries	Arteries carry blood away from the heart.
7	**(b)** water and gravity	Hydrotropism and geotropism.
8	**(d)** ovum	Egg cell.
9	**(b)** A scar from an accident	Gained in life.
10	**(d)** chlorophyll	Chlorophyll.
11	**(a)** sensory neurons	Sensory neurons.
12	**(b)** budding	Small bulges grow and detach.
13	**(d)** trait	Observable feature.
14	**(b)** small intestine	Intestinal juice.
15	**(b)** adrenal glands	Adrenal glands above kidneys.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Forebrain (1); Midbrain and hindbrain (1).	1+1
17	Tiny reproductive units (1); Covered by thick protective walls (1).	1+1
18	Thread-like structures in the nucleus (1); Made of DNA and protein, carrying genes (1).	1+1
19	Loss of water as vapour through stomata (1); From aerial parts of the plant (1).	1+1
20	Directional growth movement in response to a stimulus (1); Phototropism and geotropism (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	TT gives only T (1); Tt gives T and t (1); tt gives only t (1).	1+1+1
22	Stored in vacuoles (1); Falling leaves (1); Gums, resins, latex (1).	1+1+1
23	Prepares the body for emergency (1); Increases heart rate (1); Fight or flight (1).	1+1+1
24	Plants bear flowers and fruits earlier (1); Seedless plants can be grown (1); Plants are genetically identical to the parent (1).	1+1+1
	OR: Genes for different traits sort separately (1); This forms new combinations (1); It is seen in the dihybrid cross (1).	1+1+1
25	Holozoic: humans (1); Saprophytic: fungi (1); Parasitic: Cuscuta (1).	1+1+1
	OR: Receptor (1); Sensory neuron, spinal cord, motor neuron (1); Effector (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Xylem: water; phloem: food (1); Xylem: dead cells; phloem living (1); Xylem upward; phloem both ways (1); Xylem passive; phloem active (1); Vessels/tracheids vs sieve tubes (1).	1+1+1+1+1
	OR: Diabetes: insulin (1); Goitre: iodine/thyroxine (1); Dwarfism: growth hormone (1); Gigantism: excess (1); Treatments (1).	1+1+1+1+1
27	Pollen lands on the stigma (1); It grows a pollen tube through the style (1); The tube reaches the ovule in the ovary (1); Male gamete fuses with the female gamete (1); The zygote is formed (1).	1+1+1+1+1

	OR: Allele: alternative form of a gene (1); Homozygous: same alleles (1); Heterozygous: different alleles (1); F1: first filial generation (1); F2: second filial generation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** fruit	After fertilisation.
2	**(a)** XX	Two X.
3	**(b)** RBC	Red blood cells.
4	**(c)** auxin	Auxin.
5	**(c)** testes	The male hormone.
6	**(a)** not passed to the next generation	They do not change germ cells.
7	**(c)** oxygen	O_2.
8	**(a)** motor neurons	Motor neurons.
9	**(b)** sporangia	Sporangia are spore-bearing structures.
10	**(a)** nucleus	Chromosomes are in the nucleus.
11	**(d)** villi	Villi in small intestine.
12	**(d)** pituitary	Pituitary.
13	**(b)** seed	Fertilised ovule is a seed.
14	**(b)** sex chromosomes	X and Y.
15	**(b)** platelets	Platelets.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Fusion of male gamete and female gamete (1); Forms the zygote (1).	1+1
17	3 : 1 (2).	2
18	Carbon dioxide + water (1); gives glucose + oxygen in presence of light and chlorophyll (1).	1+1
19	A change in the environment that causes a response (1); Heat, light, touch (1).	1+1
20	Amoeba (1); Paramecium or Leishmania (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sexual: two parents; asexual: one parent (1); Sexual: gametes involved; asexual: no gametes (1); Sexual: more variation; asexual: little variation (1).	1+1+1
22	Dominant trait appears in F1 (1); Recessive trait is hidden in F1 (1); Recessive shows only when both alleles are recessive (1).	1+1+1
23	Breathing: inhale and exhale air (1); Respiration: release of energy from food in cells (1); Breathing is a physical process (1).	1+1+1
24	Goitre (1); Diabetes (1); Dwarfism (1).	1+1+1
	OR: Produce sperms (1); Secrete testosterone (1); Located in the scrotum for a lower temperature (1).	1+1+1
25	It helps survival in changed conditions (1); It allows adaptation (1); It leads to diversity (1).	1+1+1
	OR: Absorb water and minerals (1); Increase surface area (1); Present in roots (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Ingestion (1); Digestion (1); Absorption (1); Assimilation (1); Egestion (1).	1+1+1+1+1
	OR: Example: touching hot object (1); Receptor to sensory neuron (1); Spinal cord (1); Motor neuron to muscle (1); Hand withdrawn quickly (1).	1+1+1+1+1

27	A small bud forms due to repeated cell division (1); The bud grows in size (1); It develops mouth and tentacles (1); It separates from the parent (1); It lives independently (1).	1+1+1+1+1
	OR: Parents RRYY x rryy (1); F1 is RrYy round yellow (1); F2 gives four phenotypes (1); Ratio 9 : 3 : 3 : 1 (1); New combinations show independent assortment (1).	1+1+1+1+1

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